

Lab Concurrency Control

Lab 1 Experiencing Blocking (Row-Level Lock)

Blocking query:

Query	Query History	Query	Query History
1 BEGIN; 2 UPDATE film SET rental_rate = 4.99 WHERE film_id = 1; 3 4		1 BEGIN; 2 UPDATE film SET rental_rate = 5.99 WHERE film_id = 1; 3 4	
Data Output	Messages	Data Output	Messages
WARNING: there is already a transaction in progress UPDATE 1 Query returned successfully in 65 msec.			

Waiting for the query to complete...

Commit to release:

Query	Query History	Query	Query History
1 BEGIN; 2 UPDATE film SET rental_rate = 4.99 WHERE film_id = 1; 3 4 COMMIT;		1 BEGIN; 2 UPDATE film SET rental_rate = 5.99 WHERE film_id = 1; 3 4 COMMIT;	
Data Output	Messages	Data Output	Messages
WARNING: there is already a transaction in progress COMMIT Query returned successfully in 73 msec.		WARNING: there is already a transaction in progress COMMIT Query returned successfully in 62 msec.	

Lab 2: Replicating Non-repeatable Read

Read before update:

Query

Query History

1

BEGIN;

2

SELECT rental_rate FROM film WHERE film_id = 10;

Data Output

Messages

Notifications



SQL

Showing rows: 1 to 1   Page No: of 1    

	rental_rate numeric (4,2) 	
1	4.99	

Update:

Query

Query History

1

BEGIN;

2

UPDATE film SET rental_rate = 0.99 WHERE film_id = 10;

3

COMMIT;

Data Output

Messages

Notifications

COMMIT

Query returned successfully in 100 msec.

After update:

The screenshot shows a database query editor interface. At the top, there are tabs for "Query" and "Query History". The "Query" tab is active, displaying a SQL query in a text editor:

```
1 SELECT rental_rate FROM film WHERE film_id = 10;  
2 COMMIT;
```

Below the query editor, there are tabs for "Data Output", "Messages", and "Notifications". The "Data Output" tab is active, showing a table of results. The table has one column, "rental_rate", and one row with the value "0.99".

	rental_rate numeric (4,2) 🔒
1	0.99

Lab 3 Creating a Deadlock

Session A waiting:

The screenshot displays two side-by-side windows from SQL Developer. The left window, titled 'Query', shows a SQL script with three lines: `BEGIN;`, `UPDATE film SET rental_rate = 2.99 WHERE film_id = 100;`, and `UPDATE film SET rental_rate = 3.99 WHERE film_id = 200;`. Below the script, there are tabs for 'Data Output', 'Messages', and 'Notifications'. The 'Data Output' tab is selected, but it is empty. The right window also has a 'Query' tab with the same script. Below it, the 'Messages' tab is selected, showing the message: 'UPDATE 1' followed by 'Query returned successfully in 344 msec.'.

Deadlock:

Query Query History

```
1 BEGIN;
2 UPDATE film SET rental_rate = 2.99 WHERE film_id = 200;
3 UPDATE film SET rental_rate = 3.99 WHERE film_id = 100;
```

Data Output Messages Notifications

ERROR: deadlock detected
Process 27012 waits for ShareLock on transaction 1021; blocked by process 37792.
Process 37792 waits for ShareLock on transaction 1022; blocked by process 27012.

SQL state: 40P01
Detail: Process 27012 waits for ShareLock on transaction 1021; blocked by process 37792.
Process 37792 waits for ShareLock on transaction 1022; blocked by process 27012.
Hint: See server log for query details.
Context: while locking tuple (5,12) in relation "film"

Lab 4: Preventing Race Conditions

Immediate block:

Query Query History

```
1 BEGIN;
2 SELECT * FROM inventory WHERE inventory_id = 10 FOR UPDATE;
```

Data Output Messages Notifications

Showing rows: 1 to 1 of 1

	inventory_id [PK] integer	film_id smallint	store_id smallint	last_update timestamp without time zone
1	10	2	2	2006-02-15 10:09:17

Query Query History

```
1 BEGIN;
2 SELECT * FROM inventory WHERE inventory_id = 10 FOR UPDATE;
```

Data Output Messages Notifications

ERROR: current transaction is aborted, commands ignored until end of transaction block

SQL state: 25P02

After A commit:

Query Query History

```
1 BEGIN;
2 SELECT * FROM inventory WHERE inventory_id = 10 FOR UPDATE;
3
4 INSERT INTO rental (rental_date, inventory_id, customer_id, staff_id)
5 VALUES (NOW(), 10, 1, 1);
6
7 COMMIT;
```

Data Output Messages Notifications

COMMIT

Query returned successfully in 227 msec.

Query Query History

```
1 BEGIN;
2 SELECT * FROM inventory WHERE inventory_id = 10 FOR UPDATE;
3 ROLLBACK;
```

Data Output Messages Notifications

Showing rows: 1 to 1 of 1

	inventory_id [PK] integer	film_id smallint	store_id smallint	last_update timestamp without time zone
1	10	2	2	2006-02-15 10:09:17